Problem E1. Electric guitar *(20 points)*

Part A. String dimensions *(2 points)*

For length l_1, directly measured quantities and calculations:

Length l_1 and its uncertainty:

For string diameter d, directly measured quantities and calculations:

String diameter d and its uncertainty:

Part B. Resistivity *(3 points)*

Draw measurement circuit(s):

For string resistivity ρ, directly measured quantities, calculations and multimeter settings:

String resistivity ρ and its uncertainty:

Part C. String oscillations *(3 points)*

Draw measurement circuit(s):

Measurements and calculated data (you don't have to fill the entire table):

Graph: f versus l

Part D. Young modulus of the string *(4.5 points)*

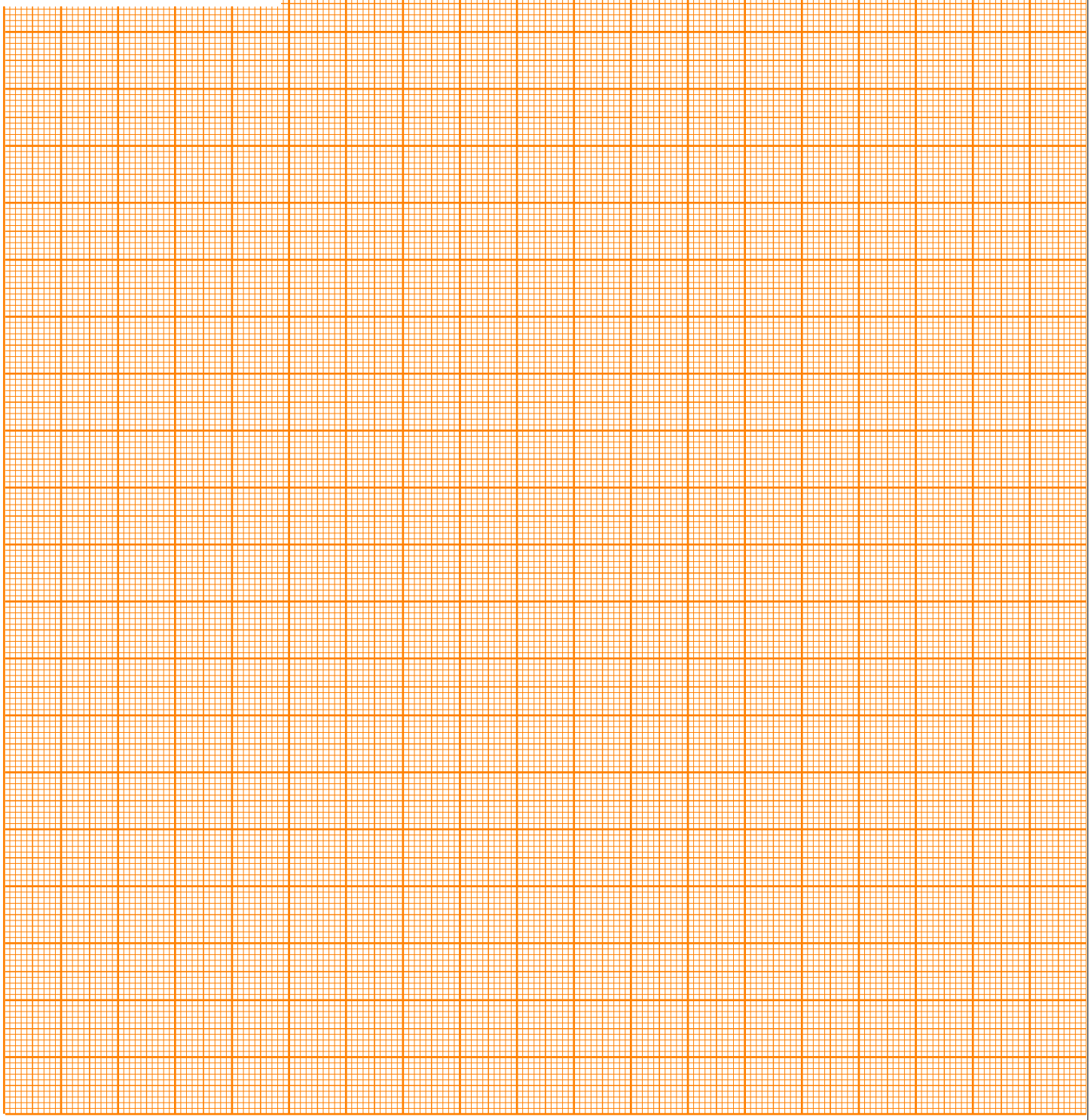
Suitable axes for finding Young modulus from graph expressed by known quantities:

x=

y=

Calculated data for the graph(you don't have to fill the entire table):

Graph: y versus x



Calculations:

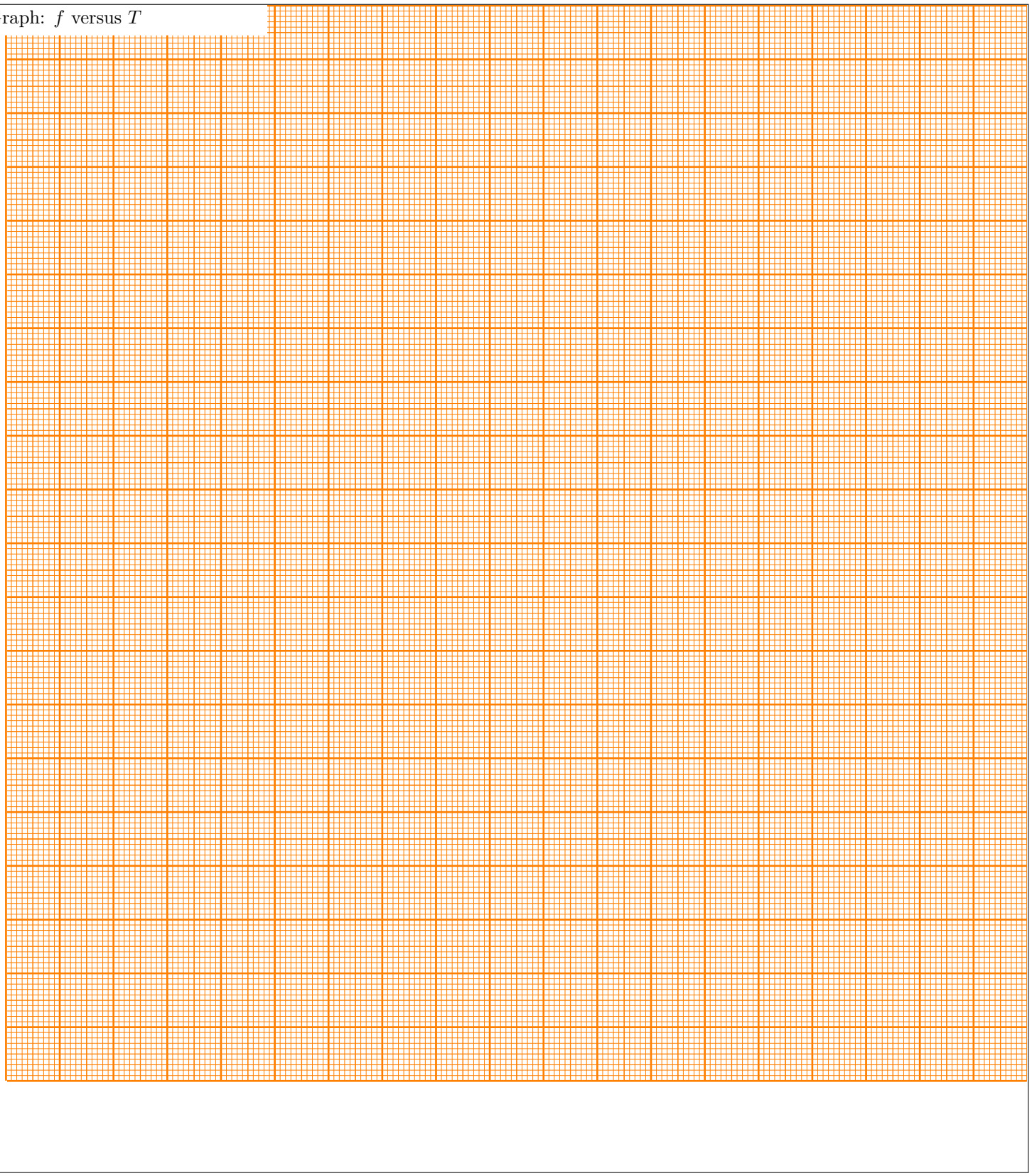
Young modulus E and its uncertainty:

Part E. Heated string *(3 points)*

Draw measurement circuit(s):

Measurements and calculated data (you don't have to fill the entire table):

Graph: f versus T



Part F. Thermal expansion of the string *(4.5 points)*

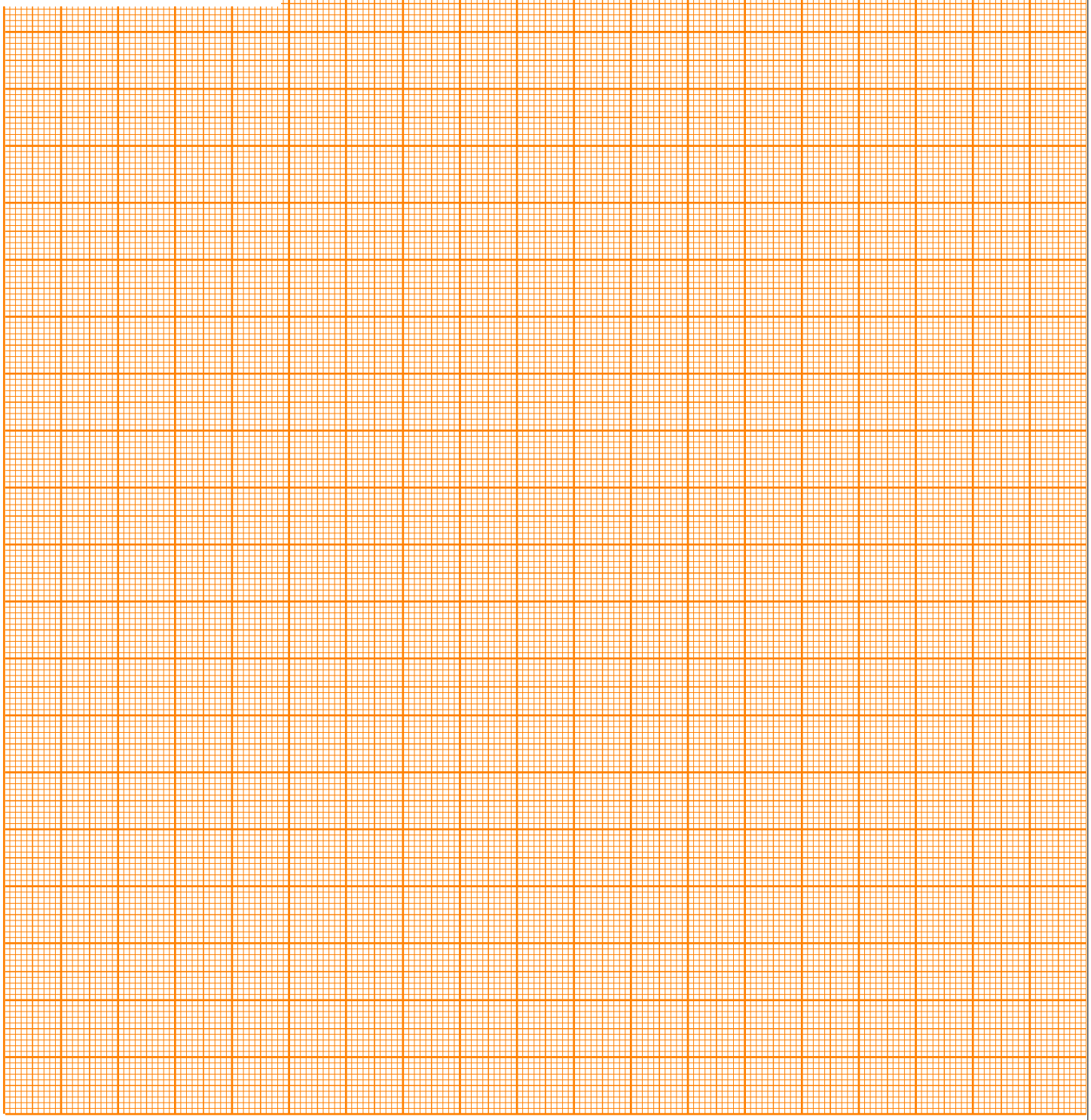
Suitable axes for finding coefficient of linear heat expansion α from graph expressed by known quantities:

x=

y=

Calculated data for the graph(you don't have to fill the entire table):

Graph: y versus x



Calculations:

coefficient of linear heat expansions α and its uncertainty: